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Crowd density as an enabling constraint: reorganization of locomotor temporal dynamics in virtual environments

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Introduction: Locomotion in everyday environments requires humans to coordinate their movements with dynamically structured social contexts such as crowds, yet little is known about how such environments influence the temporal organization of gait. This study investigated whether increasing virtual crowd density reorganizes the temporal dynamics of locomotion in community-dwelling middle-aged and older adults.

Methods: Twenty adults (aged 55–81 years) performed stepping-in-place locomotion in immersive virtual reality under three conditions: a control condition without avatars, a simple crowd with a small number of approaching avatars, and a complex crowd with higher avatar density. Step interval time series were obtained from lumbar acceleration, and locomotor dynamics were quantified using the coefficient of variation as an index of variability magnitude and the scaling exponent from detrended fluctuation analysis as an index of temporal correlations.

Results: Generalized linear mixed models showed that, relative to the no-avatar control, the complex crowd condition was associated with reduced variability magnitude and weaker temporal correlations.

Discussion: The findings suggest that socially structured virtual environments can induce a more stable yet less temporally persistent stepping pattern. In other words, successive steps became less tightly constrained by earlier fluctuations and could be adjusted more flexibly in response to the immediate context, rather than simply reflecting destabilization of stepping behavior. The findings indicate that virtual crowd environments can function as enabling constraints that reorganize locomotor temporal dynamics without compromising stability, and they highlight immersive virtual reality as a useful tool for probing and potentially training adaptive locomotor control in aging populations.

KEYWORDS

virtual reality, crowd, step variability, nonlinear dynamics, detrended fluctuation analysis

1 Introduction

Locomotion in everyday environments unfolds within dynamically structured physical and social contexts, such as crossing streets or moving through crowds. Rather than being governed solely by biomechanical or neural processes, walking emerges from continuous perception–action coupling between the individual and surrounding environmental information (Warren and Fajen, 2008). Ecological approaches to locomotion emphasize that adaptive behavior emerges from the dynamics of the interaction between a structured environment and an agent under physical and informational constraints (Warren, 2006). Empirical studies have shown that pedestrians flexibly adapt their locomotor patterns when navigating through obstacles, apertures, and other moving agents, adjusting their paths, stepping behavior and body balance in anticipation of potential collisions (Fajen, 2013; Higuchi, 2013; Olivier et al., 2012). These findings indicate that environmental and social structure provide informational constraints that shape locomotor coordination and dynamics.

In this context, the notion of *enabling environments* highlights that well-designed physical and social settings can support adaptive behavior by structuring information and constraints in ways that invite functional coordination patterns, rather than merely limiting action possibilities. Enabling environments can be understood more specifically in terms of enabling constraints, that is, environmental constraints that do not simply restrict movement but shape the conditions under which functional perception–action couplings and coordination patterns emerge in a flexible and adaptive manner. Immersive virtual reality (VR) technologies offer a promising way to create such enabling environments for locomotion by safely simulating everyday challenges, such as walking in crowds, while preserving experimental control (Ghanbari et al., 2024; De la Rosa and Breidt, 2018).

Crowd environments provide a particularly compelling example of such constraint-driven adaptation, and a large body of work has examined pedestrian behavior. Early modeling work introduced the social force model that depends on interpersonal distances and environmental layout (Helbing and Molnár, 1995; Helbing et al., 2000). However, subsequent research has pointed out that such distance-based interaction forces do not fully capture individual behavior in crowds. Moussaïd et al. (2011) proposed that crowd dynamics are better explained by incorporating visual information and simple behavioral heuristics, showing that pedestrians' local predictions about future collisions shape their avoidance behavior.

Building on this perspective, recent empirical work has focused on mutual anticipation during interpersonal interactions in crowds. Experimental studies have shown that pedestrians spontaneously coordinate their walking speed and direction when passing one another (Murakami et al., 2021; Murakami et al., 2022). Within this literature, self-organization accounts have proposed that local interactions and mutual anticipation between pedestrians give rise to emergent collective patterns such as lane formation, stop-and-go waves, and congestion at bottlenecks (Murakami et al., 2023). Most of these studies have focused on spatial and aggregate properties of movement, including trajectories, density, and flow at the group or crowd level.

By contrast, relatively little is known about how socially structured crowd environments shape the temporal dynamics of

locomotion at the level of individual walkers, particularly the structure of stride-to-stride variability in temporal aspects over time. From a dynamical systems perspective for individual performance, motor behavior emerges from the interaction of organismic, environmental, and task constraints (Newell, 1986). Within this framework, the temporal organization of locomotor fluctuations provides a key window into how adaptive/flexible a movement system is under changing demands. Age-related changes in physiological and behavioral systems have often been discussed in terms of a *loss of complexity* (Lipsitz and Goldberger, 1992; Hausdorff, 2009), but more recent work emphasizes a *loss of adaptability*, that is, a reduced ability of the system to flexibly reorganize its coordination patterns in response to changing constraints (Newell et al., 2006). Analyses of temporal variability, including its magnitude and long-range correlation structure, therefore offer an important means to examine how environmental structure and task constraints reorganize locomotor dynamics and to assess whether socially structured environments function as enabling conditions for adaptive behavior. Fractal analyses of gait variability have shown that human walking exhibits persistent temporal correlations, reflecting the adaptive organization of locomotor control (Hausdorff et al., 1997; Hausdorff, 2007), and *Detrended Fluctuation Analysis* (DFA; Peng et al., 1995) has been widely used to quantify these dynamics through the scaling exponent α , which characterizes the temporal structure of stride interval fluctuations (for methodological details, see Data Processing and Analysis in the Method section).

Recently, VR technologies have been applied to various kinds of behavioral experimental tasks including obstacle avoidance and crowd navigation (e.g., Olivier et al., 2014; Olivier et al., 2017). While real-world crowd research can approach realistic and complex situations, it is often difficult to control or manipulate experimental factors, such as the locomotor speed and direction of other pedestrians. VR technologies can provide a powerful tool for experimentally manipulating environmental factors and constraints while maintaining controlled conditions (Sato and Higuchi, 2025). For example, VR studies have experimentally manipulated environmental context and social cues to examine adaptive locomotor behavior under controlled conditions (e.g., Sharp and Cinelli, 2021; Bühler and Lamontagne, 2022). By systematically altering the number and arrangement of surrounding agents, VR environments allow researchers to examine how locomotor behavior adapts to different levels of social complexity such as crowd density (e.g., Koiliias et al., 2020). From the perspective of ecological dynamics, such environments can function as enabling constraints that reveal how perception–action systems reorganize coordination patterns under socially structured constraints.

The present study investigated how different levels of social environmental complexity influence the temporal dynamics of locomotion in a controlled VR environment. Participants performed stepping-in-place locomotion while experiencing three conditions: a control condition without avatars, a simple crowd condition with a small number of approaching avatars, and a complex crowd condition with a higher density of avatars (see Method section for details). Stepping-in-place paradigms have been widely used in virtual locomotion studies to simulate walking behavior while maintaining experimental control and

participant safety (e.g., Kondo et al., 2021). Step interval variability was quantified using the coefficient of variation (CV), where higher CV values are generally associated with less stable locomotor patterns, whereas lower values may reflect more stable or regulated stepping behavior. The temporal correlation structure of locomotor fluctuations was assessed using the DFA scaling exponent α . By manipulating social complexity (i.e., crowd density) within VR while maintaining controlled experimental conditions, this study aimed to examine how crowd-related informational constraints reorganize locomotor dynamics and to explore how socially structured environments can function as enabling conditions for adaptive coordination. We hypothesized that increasing social complexity (i.e., crowd density) would alter the temporal structure of stepping fluctuations, reflected in changes in DFA scaling exponent (α) and variability magnitude (CV).

2 Methods

2.1 Participants

Twenty community-dwelling adults (10 men and 10 women; aged 55–81 years, $M = 67.5$, $SD = 6.5$) participated in the study. We focused on middle-aged and older adults because age-related changes in temporal gait dynamics begin to progressively increase across these decades, making this population particularly relevant for understanding and training locomotor adaptability in everyday environments. All participants reported normal vision and no neurological or musculoskeletal conditions affecting gait and were able to walk independently without assistive devices. Individuals requiring walking aids (e.g., canes) were excluded.

The study was approved by the Institutional Review Board of Tokyo Metropolitan University. All procedures were conducted in accordance with institutional ethical guidelines, and written informed consent was obtained from all participants prior to participation.

2.2 Apparatus

The experimental setup included a head-mounted display (HMD; Oculus Quest 2, Meta Platforms, Inc., Menlo Park, CA, USA) and a wearable triaxial accelerometer (TSND151, ATR-Promotions, Japan), which was attached around the L4 level using a belt, positioned centrally on the lower back with the vertical axis aligned to the trunk. Acceleration data were recorded at a sampling frequency of 100 Hz. The virtual environment and experimental stimuli were developed and implemented using Unity.

2.3 Experimental stimuli

Three VR conditions were designed:

1. Control condition (No avatar): Participants wore the HMD and viewed a virtual urban street environment without avatars while stepping in place.
2. Simple condition: Approximately five avatars passed the participant every 10 s.
3. Complex condition: Approximately fifteen avatars passed the participant every 10 s.

In the control condition, only virtual street scenery was presented. In the simple and complex conditions, avatars were programmed to approach and pass in front of the participant (Figure 1). The avatars were programmed to move toward the participant at a constant speed of 4.0 km/h. The speed of the optic flow associated with the participant's stepping was individually adjusted to a comfortable walking speed prior to the experiment.

The left panel shows the simple crowd condition, with a small number of avatars approaching the participant, and the right panel shows the complex crowd condition, with a higher density of approaching avatars in the virtual street scene.

Avatars were arranged in four columns (two columns on each side), positioned approximately 1 m to the left and right of the participant. The lateral spacing between avatars was set to approximately 2.6 times the participant's shoulder width. Avatars moved at a constant speed of 4 km/h in the direction opposite to the participant's visual motion. The avatars were implemented as non-collidable visual objects in the VR environment and therefore did not produce any physical contact or haptic feedback. In addition, they were programmed to pass by participants without crossing their stepping position or requiring avoidance behavior. Consequently, no collision events occurred or were recorded during the experiment.

Optical flow of the background environment was programmed to move backward in accordance with the participant's stepping rhythm, creating the perceptual impression of forward locomotion. However, avatars moved straight ahead and were programmed not to collide with the participant.

2.4 Experimental task and procedure

Participants performed stepping-in-place on the floor. They were instructed to maintain their position and avoid forward, backward, lateral, or rotational displacement as much as possible during stepping. Although minor positional deviations may have occurred, participants were verbally guided to return to the original position when substantial displacement was observed for safety reasons. Before the main trials, the speed of the optic flow in the virtual environment was interactively adjusted while participants stepped in place wearing the HMD, so that their stepping pace matched a comfortable walking pace in the VR scene. Although the task did not involve actual forward displacement in space, the optic flow was generated to induce a sense of forward progression, making stepping-in-place a valid virtual locomotion paradigm for examining locomotor temporal dynamics. This design was adopted to maintain strict experimental control and participant safety while simulating forward locomotion with optic flow, which was particularly important given the age range of the participants (middle-aged and older adults).

Initially, they established a comfortable stepping pace. Participants then underwent a familiarization period of approximately 3–5 min, which was adjusted individually until they became comfortable with stepping in place and the VR environment. After a familiarization period with the VR environment, participants performed the stepping task under the

three experimental conditions: (1) no avatar, (2) simple, and (3) complex. The no-avatar condition was always conducted first. The order of the simple and complex conditions was counterbalanced across participants. Participants were instructed to maintain a forward gaze and continue stepping steadily during each condition. Each condition lasted approximately 5–6 min to ensure acquisition of at least 512 consecutive steps for analysis. A rest period of approximately 5 min was provided between conditions.

2.5 Data processing and analysis

Vertical acceleration data recorded from the lumbar sensor were used to detect foot contact timing. Raw acceleration signals were first smoothed using a median filter and subsequently processed with a second-order low-pass Butterworth filter (cutoff frequency: 5 Hz) to reduce noise while preserving the intrinsic temporal structure of stepping dynamics (Ravi et al., 2020). Step events were then identified from peaks in the filtered vertical acceleration signal. Detected events and waveform patterns were visually inspected to confirm signal quality and exclude obvious artifacts or misdetections.

Step intervals were derived from consecutive landing events, and step interval time series were constructed.

Variability magnitude was quantified using the coefficient of variation (CV), calculated as the standard deviation divided by the mean step interval.

Temporal structure was quantified using the scaling exponent (α) derived from Detrended Fluctuation Analysis (DFA), a nonlinear time-series method for assessing temporal correlations. DFA estimates temporal correlations by removing nonstationary trends from the time series and examining how fluctuation magnitude scales with time window size. On a log-log plot, the slope of this relationship corresponds to α , which reflects the type of temporal correlation in the signal ($\alpha \approx 0.5$: random fluctuations; $0 < \alpha < 0.5$: anti-persistent correlations; $0.5 < \alpha < 1$: persistent correlations) (see Ravi et al., 2020; Quixadá et al., 2022 for details). Changes in α have been interpreted as reflecting alterations in the adaptability/flexibility of locomotor coordination under different tasks or environmental constraints (Hausdorff, 2009; Phinyomark et al., 2020). Thus, fractal analyses of gait variability using DFA provide a useful framework for examining how environmental structure reorganizes the temporal dynamics of walking. From an ecological dynamics perspective, changes in the scaling exponent α can be interpreted not only as a loss of complexity but also as a redistribution of control across time scales, for instance from long-term persistence toward more immediate, context-sensitive regulation. DFA was applied to 512 consecutive step intervals for each condition, excluding initial and final transient steps to ensure steady-state analysis. Window sizes (4–64) were selected to increase the number of scaling points available within the 512-step time series, because other ranges (e.g., 16–N/9) yield only a limited number of points, which may reduce the stability of the regression. Candidate window ranges were compared in terms of the linearity of the log-log relationship, and the selected range showed a more stable goodness of fit than the alternative ranges.



FIGURE 1
Virtual crowd environments in the stepping-in-place task. The left panel shows the simple crowd condition, and the right panel shows the complex crowd condition.

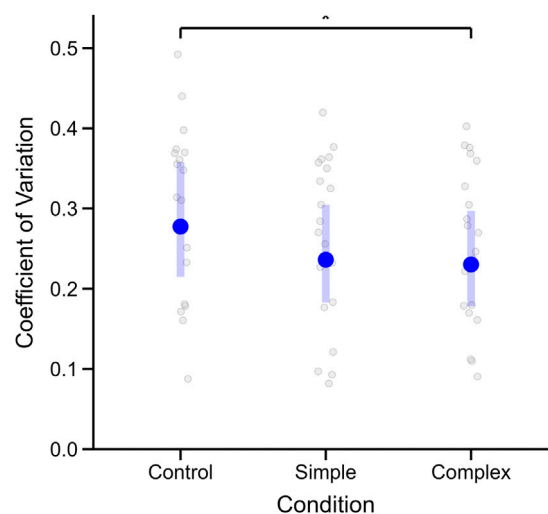


FIGURE 2
Coefficient of variation (CV) of step intervals under the three virtual crowd conditions. Gray dots represent individual participants, blue circles indicate GLMM-estimated marginal means for each condition, and shaded bands denote the 95% confidence intervals. $p < 0.05$.

Condition (categorical variables: control, simple, and complex) effects on α and CV were evaluated using generalized linear mixed models (GLMMs) with a Gamma distribution and log link. Participant was included as a random intercept. Model selection was performed by comparing linear mixed models (LMM) and GLMMs with different fixed-effect structures (Condition, Age, Sex), and model fit was evaluated using Akaike's Information Criterion (AIC). The Gamma-log GLMM with Condition as the fixed effect showed the best fit and was retained for inference. Post hoc pairwise comparisons among conditions were performed using Tukey-adjusted tests. Statistical significance was set at $p < 0.05$.

3 Results

3.1 Variability magnitude (CV)

The mean CV values were 0.297 (SD = 0.105) in the control condition, 0.261 (SD = 0.105) in the simple condition, and 0.253 (SD = 0.098) in the complex condition.

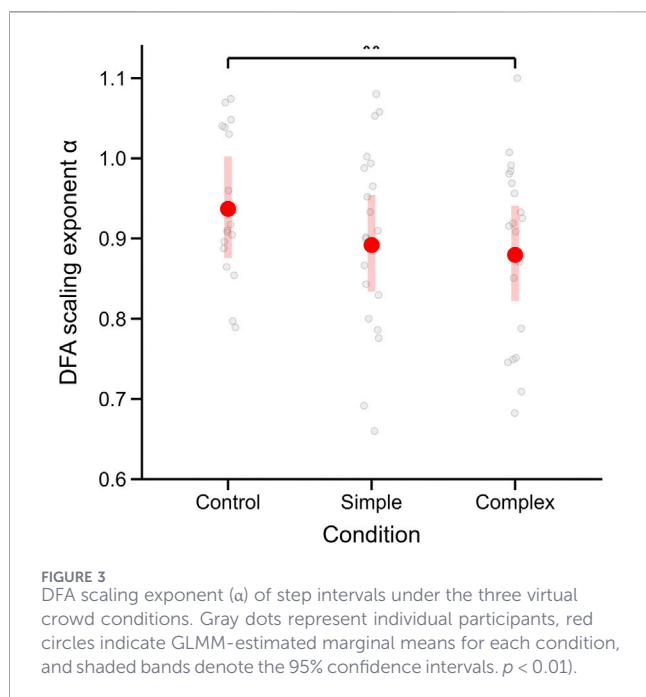


Figure 2 illustrates the magnitude of step-to-step variability (CV) across conditions. Individual participant values are shown in gray, GLMM-estimated marginal means are indicated by blue points, and shaded bands represent 95% confidence intervals. GLMM revealed a significant main effect of the condition. Post hoc pairwise comparisons with Tukey adjustment indicated that the complex condition showed significantly lower CV compared to the control condition ($p = 0.019$). The simple condition did not differ significantly from the control condition ($p = 0.052$). No significant difference was observed between the simple and complex conditions ($p = 0.928$).

These results indicate that the presence of avatars in the virtual environment was associated with reduced variability magnitude relative to the no-avatar condition, particularly under the high-density complex condition.

3.2 Temporal correlation (DFA α)

The mean α values were 0.940 (SD = 0.086) in the control condition, 0.899 (SD = 0.118) in the simple condition, and 0.887 (SD = 0.114) in the complex condition.

Figure 3 illustrates the temporal structure of step-to-step variability quantified by the DFA scaling exponent (α). Individual participant values are shown in gray, GLMM-estimated marginal means are indicated by red points, and shaded bands represent 95% confidence intervals. GLMM revealed a significant main effect of the condition. Post hoc pairwise comparisons with Tukey adjustment indicated that the complex condition showed a significantly lower α compared to the control condition ($p = 0.008$). The simple condition did not differ significantly from the control condition ($p = 0.054$). No significant difference was observed between the simple and complex conditions ($p = 0.789$).

These results indicate that the presence of avatars in the virtual environment was associated with reduced persistent temporal correlations relative to the no-avatar condition, particularly under the high-density complex condition.

4 Discussion

4.1 Reorganization of locomotor dynamics under virtual crowd conditions

The present findings indicate that socially structured virtual environments systematically reorganized locomotor dynamics in middle-aged and older adults. Both the magnitude of variability (CV) and the temporal structure of fluctuations (α) were reduced in the crowd conditions (particularly in the complex condition) relative to control. Although step detection procedures were applied carefully, part of the observed variability may reflect measurement uncertainty associated with lumbar acceleration-based step detection in addition to genuine locomotor fluctuations.

The concurrent reduction in CV and α suggests that the observed effects reflect a coordinated reorganization of locomotor control rather than simple destabilization or noise reduction. If the crowd environment merely induced instability, one would expect increased variability magnitude (CV). Instead, the results indicate a stable and less temporally persistent stepping pattern under socially structured conditions. This pattern is consistent with adaptive modulation of locomotor dynamics in response to environmental demands. One possible explanation for the reduced CV, particularly in the high-density condition, is that the presence of multiple approaching avatars may have promoted a more cautious and regulated stepping pattern. Previous studies using virtual crowd or pedestrian-avoidance paradigms have shown that gaze allocation and attention are involved in locomotor interaction with pedestrians (Meerhoff et al., 2018). Because attentional demands and cognitive load can influence gait performance (Thompson and Cinelli, 2024), we cannot completely rule out the possibility that the observed changes partly reflected visual distraction or divided attention.

On the other hand, from the perspective of the optimal movement variability hypothesis, adaptive movement is not characterized by minimal variability, but rather by an optimal amount of variability that supports both flexibility and stability (Stergiou et al., 2006). Therefore, the reduction in variability should not be interpreted simply as an improvement in locomotor control, but may also reflect a more constrained motor strategy or a reorganization of stepping dynamics whose functional meaning requires further investigation. Moreover, because perceived threat and attentional focus were not directly assessed, and because the observed variability may partly include measurement errors, these interpretations should be made with caution, and the variability indices should be interpreted carefully and further examined in future studies.

No significant difference was observed between the simple and complex crowd conditions. Although the simple condition did not differ significantly from the control condition, both avatar conditions showed lower mean CV and DFA α values than the control condition. This result may suggest that locomotor coordination may change once visually structured social cues exceed a certain threshold, which may

vary across individuals or populations. Future studies should examine a wider range of crowd densities or complexities and clinical populations to identify optimal enabling environments for therapeutic VR applications, including rehabilitation for neurological disorders such as stroke or Parkinson's disease (Khan et al., 2024; Kwon et al., 2023). The present findings further suggest that even in the absence of actual collision risk or navigation demands, visually structured crowd-like environments may influence the temporal organization of stepping behavior.

4.2 Temporal persistence and adaptive locomotor control

The decrease in DFA scaling exponent (α) provides insight into how locomotor coordination was reorganized. In stride interval dynamics, α values near 1.0 are typically interpreted as reflecting persistent correlations across time scales. A reduction in α indicates weaker temporal persistence, suggesting that stepping intervals became less constrained by previous fluctuations over extended time horizons.

Such a shift may reflect redistribution of control across time scales. Rather than relying on strongly persistent temporal patterns, participants may have engaged in more immediate, context-sensitive regulation of stepping rhythm. In socially structured VR environments, even without actual physical avoidance, the perceptual presence of approaching avatars may be associated with changes in attentional allocation or anticipatory adjustments, consistent with studies showing that virtual humans' approach, gaze and facial expression modulate interpersonal distance and personal space regulation (Bailenson et al., 2003; Ruggiero et al., 2021). This could reduce persistent temporal dependence while maintaining overall stability.

A comparable pattern has been reported in studies of locomotor expertise. Nakayama et al. (2010), examining stride interval dynamics during treadmill running, compared long-distance expert runners with novice runners. They found that the expert runners exhibited lower CV and lower DFA scaling exponents (α). This pattern was interpreted as indicating that skilled runners are able to suppress variability in running rhythm without losing the degrees of freedom necessary for flexible spatiotemporal organization of gait pattern (Nakayama et al., 2010). Although the mechanisms underlying expertise-related changes in running and acute adaptations to VR crowds are likely not identical, a conceptually similar pattern may be observed in the present study: the visually complex crowd environment may have induced an adaptive state in which participants reduced the magnitude of stepping variability while maintaining sufficient degrees of freedom in the locomotor system to flexibly adjust their movements. In this sense, the presence of visually structured social agents may have promoted a preparatory locomotor state that balances stability with adaptability, allowing participants to remain ready to respond to potential interactions with surrounding agents.

Importantly, this interpretation is consistent with the concurrent reduction in CV. The decrease in α did not coincide with increased variability magnitude, suggesting that locomotor behavior did not become more unstable. Instead, the combined changes in linear (CV) and nonlinear (α) indices indicate a

reorganization of locomotor coordination across time scales. In particular, the pattern of reduced CV together with reduced α points to a control state that is both globally stable and more easily adjusted at shorter time scales.

4.3 Virtual crowds as enabling environments for locomotion

These findings contribute to the concept of enabling environments by showing that visually complex crowd environments can alter locomotor dynamics without compromising stability. Rather than impairing gait control, the presence of moving avatars appears to act as an enabling constraint that reshapes coordination patterns. Notably, the strongest effect emerged under the high-density complex condition, where variability magnitude was significantly reduced relative to the no-avatar condition. This suggests that increased environmental complexity may promote adaptive adjustments in locomotor coordination, encouraging a more organized and responsive stepping pattern.

Recent work in real crowds similarly suggests that allowing pedestrians to maintain flexible, asynchronous stepping can support robust spatial self-organization, whereas constraining their footstep timing can impair collective pattern formation (Tomaru et al., 2024). At the same time, previous studies have also shown that fractal auditory and visual metronomes can modify the fractal dynamics of walking, altering stride-to-stride fluctuations and their long-range correlations (Marmelat et al., 2014; Rhea et al., 2014). In contrast, the present study demonstrates that socially structured visual environments in VR—rather than explicit rhythmic cues—can reorganize locomotor control by jointly changing variability magnitude (CV) and the DFA scaling exponent (α). By introducing structured social information in a safe and controlled manner, VR environments can therefore facilitate adaptive changes in the temporal organization of gait.

Consistent with this interpretation, environmental complexity should not necessarily be viewed as a perturbation that disrupts movement. Instead, the present results suggest that structured social contexts may facilitate refinement or reconfiguration of locomotor dynamics. Even in a stepping-in-place paradigm without real navigation, the presence of socially meaningful agents influenced temporal organization. This underscores the role of perceived environmental structure in shaping movement coordination patterns.

From an applied perspective, socially structured VR environments may offer a promising platform for designing contexts that stimulate adaptive locomotor control in aging populations, for example, by safely training walking in crowded public spaces or transport hubs. Future research should examine whether repeated exposure to such environments enhances functional adaptability during overground walking and whether similar reorganizational effects are observed in populations with mobility impairments.

4.4 Limitations and future directions

Several limitations should be acknowledged.

First, the present study employed stepping in place within a VR environment rather than overground walking in a real-world context. In particular, stepping in place differs from real walking with forward displacement in terms of the need to control dynamic balance during continuous progression and the availability of vestibular cues generated by actual translational self-motion (Campos and Bühlhoff, 2012; Janeh and Steinicke, 2021). These differences may influence locomotor regulation and should be considered when interpreting the present findings. Although the VR setup allows precise manipulation of social complexity under controlled conditions, ecological validity may be limited. Future research should examine whether similar reorganizational effects emerge during natural locomotion in the real-world.

Second, the sample comprised a modest number of community-dwelling middle-aged and older adults ($n = 20$), which is comparable to previous VR locomotion studies but nevertheless limits the generalizability of the present findings. Approximate post-hoc power analyses suggested moderate to acceptable statistical power for the primary effects, although larger samples remain warranted to improve robustness. Future work should replicate and extend these effects in larger and more diverse samples. Because the present sample consisted primarily of middle-aged and older adults, the observed effects should be interpreted in relation to this age group. Aging is known to influence gait variability and locomotor temporal dynamics (e.g., Hausdorff et al., 1997; Ducharme and van Emmerik, 2018), and therefore the present findings may not generalize to younger adults or other age groups. In addition, fall history was not assessed in the present study, and future studies should examine whether prior falls influence locomotion outcomes to crowd-like VR environments. It will be important to examine whether similar reorganizational effects are observed in clinical populations with gait and balance impairments, particularly neurological populations such as individuals with stroke or Parkinson's disease, because different impairments may involve distinct mechanisms of locomotor adaptation (e.g., Villeneuve et al., 2024; De Aquino Costa Sousa et al., 2024).

Third, regarding the experimental design, the control condition was always conducted first because it served as the baseline condition before participants were exposed to avatar conditions, whereas the subsequent avatar conditions were counterbalanced. However, possible order effects, including habituation to the VR environment or stepping task, cannot be completely ruled out and may have influenced the observed indices. Therefore, future studies should adopt designs that further minimize potential order effects. In addition, although participants did not report substantial fatigue during the experiment, objective analyses of fatigue or habituation effects were not conducted. Future studies should examine fatigue or habituation effects using objective temporal analyses.

Finally, a limitation of the present study is the simplicity of the avatars' behavior and appearance in the virtual environment. In the current study, environmental complexity was manipulated through crowd density, operationalized by the number and spatial density of avatars passing within a given period. However, this density manipulation was relatively regular and simplified, because the avatars moved straight ahead at a constant speed. Under such conditions, avatars passed by the participants without posing any risk of collision, crossing their path, or engaging in direct interaction.

In addition, avatars positioned in the outermost columns may have been less noticeable to participants and may have contributed less to perceived social threat than centrally located avatars. Nevertheless, the presence of these avatars and such a spatial crowd density still influenced the temporal structure of participants' stepping behavior. This finding is noteworthy, as it suggests that even minimal social cues can modulate locomotor dynamics. Future studies should examine more complex and socially rich avatar configurations, such as agents approaching from oblique directions, exhibiting varied appearance or mobility constraints (e.g., Bourgaize et al., 2024; Thompson and Cinelli, 2024; Nikmanesh et al., 2025). In natural crowd environments, locomotion is shaped not only by the presence of others but also by mutual anticipation and interaction among pedestrians (Murakami et al., 2021; Moussaïd et al., 2011). Pedestrians continuously adjust their movements based on predictions of others' future trajectories. In addition, examining where participants direct their gaze and how visual information supports anticipatory adjustments may provide deeper insight into the perceptual mechanisms underlying locomotor adaptation in socially structured environments (Higuchi, 2013; Berton et al., 2020; Meerhoff et al., 2018).

In conclusion, despite these limitations, the controlled manipulation of social complexity within VR offers a promising framework for understanding how environmental design can enable adaptive reorganization of locomotor coordination. Taken together, the present findings suggest that socially structured virtual crowd environments can be leveraged as enabling constraints to probe and potentially train adaptive locomotor dynamics.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

The studies involving humans were approved by the Ethics Committee of Tokyo Metropolitan University. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

Author contributions

KK: Conceptualization, Data curation, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review and editing. KS: Data curation, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review and editing. RS: Conceptualization, Investigation, Methodology, Supervision, Validation, Writing – original draft, Writing – review and editing. TT: Methodology, Validation, Writing – original draft, Writing – review and editing. TH: Funding acquisition, Project

administration, Validation, Writing – original draft, Writing – review and editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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